

Announcements

- ▶ Any questions/issues?

- ▶ Schedule:

Today (Follow-up): Discussion Q2 (estimating solar constant)

Today (New): Atmosphere and Greenhouse Effect (week 2) — Electricity and Magnetism, and Quantum Mechanics!

Tomorrow (Lab): Cooling by radiation, interaction of light and molecules

Thursday (Discussion): Understanding molecule-light interactions, spectral lines of atmosphere

- ▶ Next Week: Water and Phase transitions (Part I)

Today's Reading Quiz (3 min)

On your own sheet of paper...

What makes a particular gas a greenhouse gas? Give two examples of greenhouse gases and two examples of atmospheric gases that are not greenhouse gases.

Turn in to Blackboard as scanned pdf (or image is fine).

Question 2 from Discussion

(b) Below is a table of data taken for a [class from University of Arizona](#) (where the Sun is a more reliable partner). Let's try to estimate the solar constant³, S , using the data.

- Using graph paper, plot this data (Temperature (C) vs. time (s)) being careful to set the range on each axis so that no space is wasted (look at the minimum and maximum values of each quantity).
- Exam the data and choose a range of times where the temperature seems to be increasing approximately linearly (away from the initial and final "flatness").

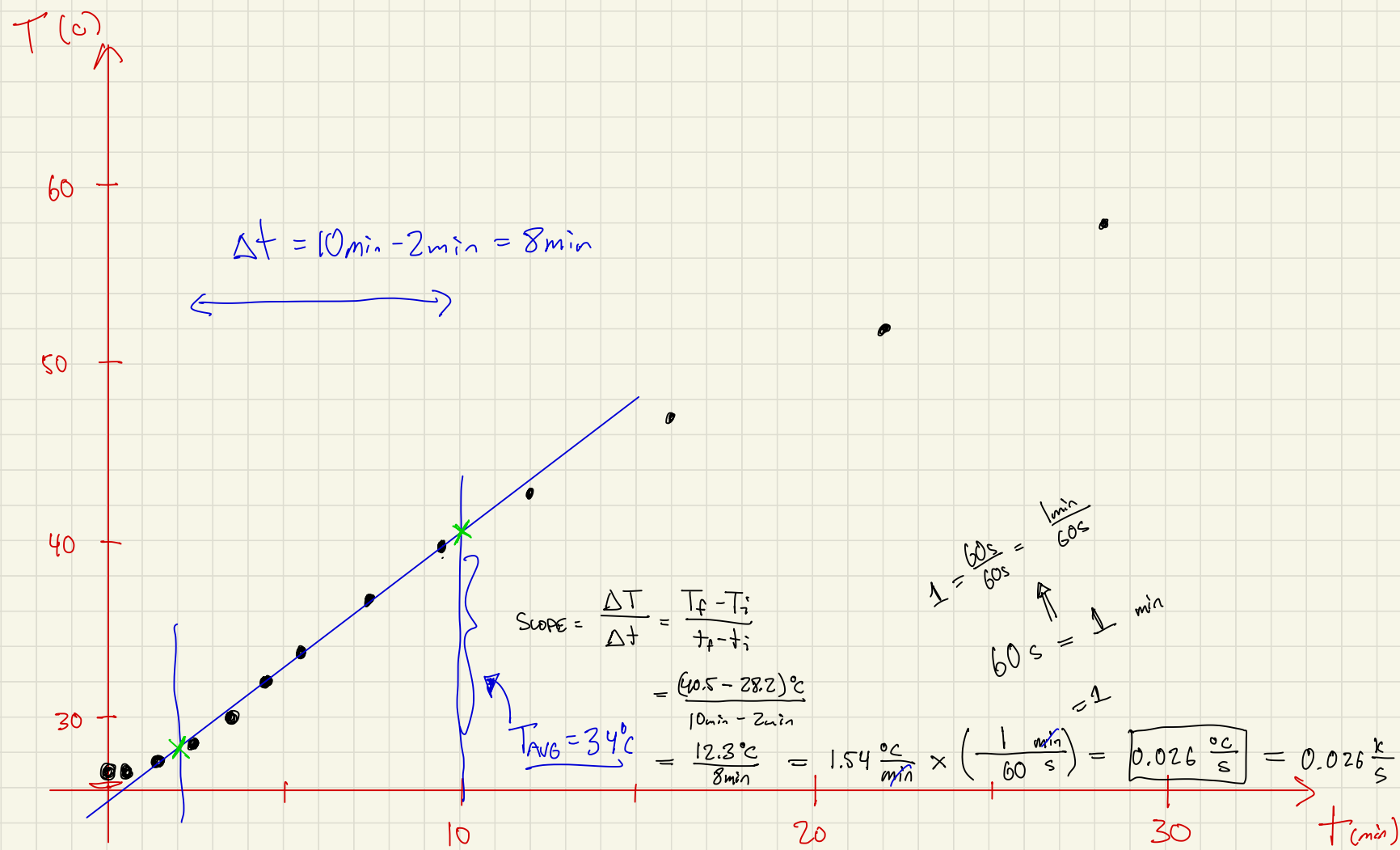
Aluminum block dimensions: 7.0 cm x 13.4 cm (93.8 cm³)
 Mass of aluminum block: 324 grams
 Specific heat of aluminum: 0.215 cal/(gram °C)

Time	Elapsed Time	Temperature
9:09:30	0.0 min	27 C (80.6 F)
9:10:00	0.5	27 (80.6)
9:11:00	1.5	27.5 (81.5)
9:12:00	2.5	28.5 (83.3)
9:13:00	3.5	30.0 (86.0)
9:14:00	4.5	32.0 (89.6)
9:15:00	5.5	33.5 (92.3)
9:17:00	7.5	36.5 (97.7)
9:19:00	9.5	39.5 (103.1)
9:21:30	12.0	42.5 (108.5)
9:25:30	16.0	47.0 (116.6)
9:31:30	22.0	52.0 (125.6)
9:38:00	28.5	56.0 (132.8)

27°C → 56°C
 (VERT AXIS)

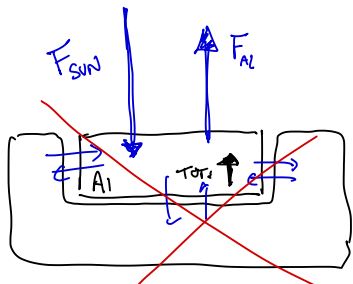
0 - 30 min (HORIZ AXIS)

- What is the average temperature of the aluminum during that time interval?
- Use a ruler to draw a line of best fit through this portion of the data (you can draw the straight line across your whole graph, but only fit to your chosen data points).
- Find the slope of that line and its units (always use two widely-separated points on the line to find a slope, not data points).
- The slope you found is $\frac{\Delta T}{\Delta t}$ for the aluminum block. Now use the defining equation for specific heat capacity and the specific heat of aluminum (0.9 J/g/K) to determine the power gain of the aluminum, $\frac{\Delta E}{\Delta t}$.



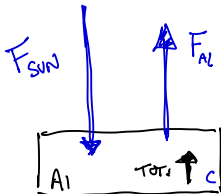
Question 2 from Discussion

$$\frac{\Delta T}{\Delta t} = 0.026 \frac{^{\circ}\text{C}}{\text{s}}$$



~~DON'T NEED STYROFOAM~~

$$\begin{aligned} A_{A1} &= 7\text{cm} \times 13.4\text{cm} = 93.8\text{cm}^2 \\ &= 93.8\cancel{\text{cm}^2} \times \frac{1\text{m}}{100\cancel{\text{cm}}} \times \frac{1\text{m}}{100\cancel{\text{cm}}} \\ &= (9.38 \times 10^1) \times 10^{-2} \times 10^{-2} \text{m}^2 \\ &= 9.38 \times 10^{-3} \text{m}^2 = 0.00938 \text{m}^2 \end{aligned}$$



IN ENERGY:

$$\Delta E = Q + W$$

$$\Delta E = Q_{\text{sun}} + Q_{A1}$$

$$\frac{\Delta E}{\Delta t} = \frac{Q_{\text{sun}}}{\Delta t} + \frac{Q_{A1}}{\Delta t}$$

$$\frac{\Delta E}{\Delta t} = + F_{\text{sun}} \cdot A_{A1} - F_{A1} \cdot A_{A1}$$

($\frac{\text{J}}{\text{s} \cdot \text{m}^2}$)

$$\left. \begin{aligned} \frac{\Delta E}{\Delta t} &= F_{\text{sun}} A - F_{\text{Ar}} A \\ \frac{\Delta T}{\Delta t} &\rightarrow \frac{\Delta E}{\Delta t} \end{aligned} \right\}$$

$$m_{\text{Al}} c_{\text{Al}} \frac{\Delta T}{\Delta t} = F_{\text{sun}} A - \left(6 \times 10^{-8} \frac{\text{J/s}}{\text{m}^2 \text{K}^4} T_{\text{Al}}^4 \right) A$$

$$\frac{m_{\text{Al}} c_{\text{Al}} \frac{\Delta T}{\Delta t}}{A} = F_{\text{sun}} - \left(6 \times 10^{-8} T^4 \right)$$

$$F_{\text{sun}} = \frac{m_{\text{Al}} c_{\text{Al}} \frac{\Delta T}{\Delta t}}{A} + 6 \times 10^{-8} T^4$$

$$= \frac{7.5 \text{ J/s}}{0.01 \text{ m}^2} + 6 \times 10^{-8} \cdot (37 + 273)^4$$

$$= 750 \frac{\text{J/s}}{\text{m}^2} + 530 \frac{\text{J/s}}{\text{m}^2}$$

$$= 1280 \frac{\text{J/s}}{\text{m}^2}$$

$$S = 1360 \frac{\text{W}}{\text{m}^2}$$

$$\Delta E = m_{\text{Al}} c_{\text{Al}} \Delta T \leftarrow$$

$$\Delta T = \frac{\Delta E}{m_{\text{Al}} c_{\text{Al}}}$$

$$\frac{\Delta T}{\Delta t} = \frac{\Delta E}{m_{\text{Al}} c_{\text{Al}} \Delta t}$$

$$\frac{\Delta E}{\Delta t} = m_{\text{Al}} c_{\text{Al}} \left(\frac{\Delta T}{\Delta t} \right)$$

$$= (3273) \left(0.9 \frac{\text{K}}{\text{g}} \right) \frac{\Delta T}{\Delta t}$$

$$= (3273)(0.9)(0.026) \frac{\text{J/s}}{\text{J/s}}$$

$$= 7.5 \text{ J/s}$$

Charge and Electric Force

- ▶ There are two types of electric charge: positive (held by protons) and negative (held by electrons).
- ▶ Atoms are neutral (same number of protons and electrons), but could become ionized (lose or gain electron) or shift their distribution of charge when bonding.
- ▶ Like charges repel, opposite charges attract. (the *Coulomb Force*)

Electric Field

- ▶ Physicists don't like “action at a distance” so they use *fields* produced by charges (or masses: the gravitational field) to convey forces.
- ▶ Positive charges have electric fields pointing away, negative have fields pointing towards.

Force of electric field on charge: $\vec{F}_E = q\vec{E}$ (opp/same dir $q \lessgtr 0$)

Electric Dipole

A perfect dipole is two equal magnitude charges held apart at a distance d . We write it as a vector $\vec{\mu}$ (Greek letter μ)

Electromagnetic Wave and a dipole

Interaction with EM field and a dipole (Classical)

Interaction with EM field and a dipole (Quantum)

Chemistry: Building Molecules

- ▶ In Physics we construct the energy levels of a Hydrogen atom
- ▶ The periodic table (Chemistry) fills in those energy levels with electrons ($1 = \text{H}$, $2 = \text{He}$, $3 = \text{Li}$, ...)
- ▶ Recall the “octet rule” and Dot/Lewis diagrams

Chemistry: Building Molecules

- ▶ Atmospheric gases that are not greenhouse, have no dipole.

Chemistry: Building Molecules

- ▶ Atmospheric gases that are greenhouse, have the *possibility* of a dipole.